AI-Driven Housing Affordability Forecasting in New York City:

An NTA-Level Panel Analysis Using Ensemble Machine Learning

Abstract

Housing affordability is a defining governance challenge in New York City (NYC), where over 52% of renter households are cost-burdened and approximately 30% are severely cost-burdened, spending more than 50% of gross income on housing. This study develops a neighborhood-scale, explainable machine-learning framework to forecast housing affordability at the Neighborhood Tabulation Area (NTA) level using a panel of 2,512 observations across 239 NTAs from 2012–2022. We integrate 49 socio-economic, housing market, and rent indices features from ACS 5-year estimates, eviction records, and the Zillow Observed Rent Index (ZORI). Three gradient-boosted ensemble models—Random Forest, XGBoost, and LightGBM—are benchmarked under a strict temporal train/validation/test split with 5-fold TimeSeriesSplit cross-validation to prevent data leakage. LightGBM achieved the highest predictive performance (Test R^2=0.926, RMSE=0.035, MAE=0.019). SHAP TreeExplainer analysis identifies rent burden, vacancy rate, and rent-to-income ratio as dominant drivers, confirming that NYC's rent escalation follows a systemic "burden escalator" across boroughs. Spatial diagnostics (Moran's I) show no significant autocorrelation in residuals, and ablation analysis confirms rental market variables drive the largest incremental contribution. Based on forecasts, we propose a 14-instrument policy matrix including early-warning interventions, borough-differentiated zoning and voucher strategies, and anticipatory resource allocation. This framework demonstrates how explainable AI can operationalize predictive insights into actionable urban governance strategies, offering a replicable model for global cities facing housing crises.

Index Terms-*housing affordability, machine learning, ensemble methods, XGBoost, LightGBM, SHAP explainability, New York City, neighborhood tabulation area, temporal cross-validation, spatial autocorrelation, rent burden, urban informatics, panel data, housing policy, early-warning system, borough-level forecasting, rent burden escalation, policy recommendation matrix*

I. Introduction

New York City (NYC) faces the United States' most severe housing affordability crisis. As of the 2022 American Community Survey (ACS), over 52% of renter households are cost-burdened—spending more than 30% of gross income on housing—and approximately 30% are severely cost-burdened, exceeding the 50% threshold set by the U.S. Department of Housing and Urban Development (HUD). Severe rent burden forces households to trade housing security against basic necessities such as food, healthcare, and education. Rental costs in NYC have consistently outpaced renter incomes across every borough since at least the 1980s, with the gap widening sharply during and after the COVID-19 pandemic. Addressing this crisis requires not only accurate measurement but actionable, neighborhood-level intelligence to guide housing assistance resources toward the highest-risk communities before administrative data confirms deterioration.

The Neighborhood Tabulation Area (NTA) is the official sub-borough geographic unit designated by the NYC Department of City Planning (DCP), comprising approximately 15,000–40,000 residents per neighborhood. With 263 NTAs spanning four boroughs (Brooklyn, Queens, Manhattan, and the Bronx), this unit captures meaningful variation in housing market conditions while maintaining statistically reliable ACS sample sizes, making it the natural unit for neighborhood-targeted housing policy. Staten Island is excluded from this study due to limited ACS coverage and smaller sample reliability at the NTA level. However, excluding Staten Island does not affect the generalizability of results to the majority of NYC's rental population. Most existing machine learning applications to NYC housing operate at the borough or metropolitan level, leaving NTA-level panel structures and neighborhood-level policy targeting largely unexplored.

This paper addresses this gap through five primary contributions:

1. **Data construction:** We create a novel NTA-level panel dataset of 2,512 observations across 239 NTAs in four boroughs (2012–2022), integrating ACS 5-year estimates, NYC eviction court records, and the Zillow Observed Rent Index (ZORI).
2. **Model benchmarking:** We rigorously evaluate three gradient-boosted ensemble models—Random Forest, XGBoost, and LightGBM—under a strict temporal train/validation/test split with 5-fold TimeSeriesSplit cross-validation to prevent data leakage.

3. **Explainable ML analysis:** Using SHAP TreeExplainer, we generate global and borough-level feature importance rankings, including the first borough-comparative SHAP decomposition showing that severe rent burden follows a uniform structural escalation mechanism across all four boroughs.

4. **Spatial diagnostics and forecasting:** We conduct Moran's I analysis on model residuals and produce three-year borough-level affordability forecasts (2023–2025) under a calibrated baseline macroeconomic scenario.

5. **Policy translation:** We propose a 14-instrument policy recommendation matrix, anchored to SHAP attributions, forecast trajectories, and model precision evidence, identifying early intervention at the 30% burden threshold, borough-differentiated supply and income-side responses, and NTA-level predictive targeting to operationalize anticipatory housing policy in NYC.

The remainder of this paper is organized as follows. Section II reviews housing affordability and machine learning literature. Section III describes dataset construction. Section IV details feature engineering. Section V presents the methodological pipeline. Section VI reports experimental results. Sections VII, VIII, and IX present SHAP analysis, ablation study, and spatial autocorrelation results. Section X provides borough-level forecasts. Section XI discusses policy implications and the recommendation matrix. Section XII addresses limitations and future directions. Section XIII concludes.

II. Related Work

A. Housing Affordability: Measurement and Theory

The 30%-of-income threshold for housing affordability was codified in the National Housing Act of 1937 and subsequently reinforced through HUD administrative rules governing federal housing assistance eligibility. Stone [2] challenged this fixed-ratio approach with the shelter poverty framework, arguing that affordability must be evaluated against the residual income available for non-housing necessities after rent is paid -a standard that is more stringent for low-income households and more permissive for high-income households than the uniform 30% threshold. Quigley and Raphael [3] provided rigorous empirical evidence that restrictive land-use regulation is the primary structural driver of affordability deterioration in coastal cities including NYC, establishing the supply-constraint mechanism that underlies the rent growth dynamics modeled in this paper. Glaeser and Gyourko [4] estimated that zoning regulations impose implicit welfare costs equivalent to 50% of market rents in highly regulated cities, directly reducing housing supply elasticity and preventing the market-clearing response to demand increases. More recently, Albouy, Ehrlich, and Liu [23] demonstrated that place-based sorting amplifies affordability inequality: high-productivity workers outbid lower-income renters in desirable neighborhoods, concentrating severe burden in spatially identifiable clusters -a dynamic that motivates the NTA-level geographic disaggregation central to this analysis. Collectively, this literature establishes that severe rent burden is generated by the intersection of supply constraints, income inequality, and spatial sorting, all of which are captured in the feature set constructed in Section IV.

B. Ensemble Machine Learning for Housing Prediction

Gradient-boosted decision tree ensembles have emerged as the leading machine learning paradigm for property valuation and housing market prediction, consistently outperforming hedonic regression and neural network alternatives on structured tabular data. Antipov and Pokryshevskaya [5] demonstrated that gradient boosting outperforms hedonic regression for mass appraisal, achieving R^2=0.89 on Russian residential data. Park and Bae [6] applied Random Forests to Korean residential price forecasting, finding that ensemble methods substantially outperform multiple regression on out-of-sample data. Chen and Guestrin [13] introduced XGBoost, demonstrating its scalability and regularization advantages through the regularized objective $L(\varphi) = \Sigma_i\, l(y_i, \hat{y}_i) + \Sigma_k\, [\gamma T_k + \frac{1}{2}\lambda \|w_k\|^2]$, where γ penalizes tree complexity and λ provides L2 weight regularization. Ke et al. [14] introduced LightGBM with histogram-based leaf-wise tree growth, achieving substantially faster training on high-dimensional tabular data while matching or exceeding XGBoost accuracy on benchmark datasets -a performance advantage confirmed in the held-out test results in Section VI. In the NYC housing context, Been, Ellen, and Madar [7] applied logistic regression to building-level eviction risk using administrative records, but without ensemble methods, NTA panel structure, or temporal cross-validation. This paper extends the ensemble ML literature to the specific context of NTA-level severe rent burden prediction.

C. SHAP Explainability in Applied Research

Lundberg and Lee [8] introduced SHAP (SHapley Additive exPlanations) as a theoretically grounded feature attribution framework rooted in cooperative game theory, satisfying the axioms of local accuracy, missingness, and consistency that prior feature importance methods fail to jointly satisfy. TreeExplainer provides exact polynomial-time Shapley values for tree ensembles, enabling computationally efficient global and local explanations. The global importance measure $\bar{I}_j = (1/n)\Sigma_i|\varphi_j(i)|$

aggregates directionally-neutral marginal contributions across observations. Abidoye and Chan [9] applied SHAP to residential valuation in Hong Kong, demonstrating the utility of SHAP for policy-relevant interpretation of housing market models. Molnar [24] provides a comprehensive treatment of SHAP's theoretical properties, practical interpretation guidelines, and limitations in the presence of highly correlated features -a consideration directly relevant to the multicollinearity between rental market features identified in the ablation study of Section VIII. This paper presents the first application of SHAP TreeExplainer to NTA-level rent burden prediction in NYC, extending prior work by providing both global feature rankings and borough-level comparative decompositions.

D. Spatial Econometrics and Urban Panel Data

Anselin [10] formalized spatial dependence diagnostics through Moran's I and spatial lag and error regression models, establishing the methodological framework applied in Section IX. LeSage and Pace [11] demonstrated that ignoring spatial autocorrelation in panel residuals produces inconsistent parameter estimates under omitted spatially correlated confounders. Gibbons and Machin [16] showed strong positive spatial autocorrelation in London housing price gradients, consistent with neighborhood amenity spillovers that may operate at the sub-borough NTA level in NYC. In temporal panel contexts, Case and Shiller [17] demonstrated the importance of controlling for temporally persistent unobserved heterogeneity in housing panel data, motivating the temporal lag features constructed in Section IV.C. James et al. [18] provide the foundational treatment of cross-validation bias in temporal settings: standard k-fold cross-validation inappropriately uses future data to predict the past, requiring temporal splits or TimeSeriesSplit protocols -the precise rationale for the strict temporal partition and 5-fold TimeSeriesSplit cross-validation protocol described in Section V.

E. Machine Learning for Housing Policy Targeting

An emerging strand of applied machine learning research has moved beyond predictive accuracy toward policy-operational deployment of housing market models. Desmond and Gershenson [25] demonstrated that eviction risk prediction models constructed from administrative court records can identify high-risk households with sufficient precision for targeted pre-eviction legal assistance programs, establishing the precedent for operationalizing predictive models as policy-targeting instruments. Glaeser et al. [26] applied machine learning to identify building-level housing code violation risk in Boston, demonstrating that model predictions can guide proactive inspection prioritization with measurable improvements in violation detection rates. These applications share a common structure with the early-warning system proposed in Section XI.E: a predictive model trained on administrative data generates neighborhood-level risk scores that allow housing agencies to pre-position resources ahead of crisis manifestation. However, no prior study has applied this framework to NTA-level severe rent burden prediction in NYC, integrated SHAP attribution with borough-level forecasting to produce policy-differentiated recommendations, or demonstrated the temporal advantage of model-based predictions over ACS-lagged administrative burden statistics.

F. Gap in the Literature

Despite substantial prior work, four gaps remain:

1. No study applies ensemble ML to NTA-level panel data in NYC over a decade, leaving neighborhood-level spatial structure unexplored.
2. No prior NYC study integrates SHAP attribution, temporal cross-validation, leave-one-group-out ablation, and spatial autocorrelation diagnostics in a single analytical pipeline.
3. Borough-level forecasting of severe rent burden under explicit macroeconomic scenarios has not been documented, leaving planners without predictive insights.
4. NTA-level ML outputs have not been translated into structured, empirically anchored policy recommendation matrices that link SHAP results and forecast trajectories to actionable instruments, responsible agencies, and urgency levels.

This study addresses these gaps, providing robust, interpretable, and policy-relevant insights for anticipatory urban governance.

III. Dataset Construction

A. Primary Data Sources

The dataset integrates three primary sources, each providing distinct and complementary dimensions of neighborhood housing market conditions across the 2012–2022 study period. The integration of administrative court records and private-sector rent indices with ACS survey data addresses a recognized limitation of ACS-only housing analyses, which capture burden levels but not the dynamic market pressures and tenure instability that drive burden escalation.

1) American Community Survey 5-Year Estimates (2012–2022):

Twenty-three socioeconomic and housing variables are obtained at the census tract level via the U.S. Census Bureau Data API for ACS 5-year estimate releases covering reference years 2012 through 2022. ACS 5-year estimates pool five consecutive years of survey responses to produce statistically reliable estimates for small geographic units, making them the appropriate data product for NTA-level analysis where single-year estimates would carry prohibitively large margins of error for tracts with fewer than 2,000 renter-occupied households. Key ACS tables utilized include B19013 (median household income), B25119 (renter median income by tenure), B25064 (median gross rent), B25060 (median contract rent), B25070 (gross rent as a percentage of household income, providing both the 30% and 50% burden threshold counts), B25002 (housing unit vacancy status), B25014 (rooms per unit by tenure, enabling severe crowding rate computation), B19083 (Gini coefficient of income inequality), and B08301 (commuting mode, enabling transit commute share computation). All variables are obtained at the census tract level and subsequently aggregated to NTAs using the population-weighted procedure described in Section III.B. ACS 5-year estimates for a given reference year are released with an 18-to-24-month publication lag -a constraint with direct implications for the operational early-warning system proposed in Section XI.E.

2) NYC Eviction Court Records (2012–2022):

A total of 91,198 residential eviction filing records are obtained from the NYC Office of Court Administration through NYC Open Data (dataset identifier: 6z8x-wfsh), covering all residential eviction proceedings filed in NYC Housing Court during the 2012–2022 study period. Records are geocoded to census tracts using building-level addresses and spatially aggregated to NTA-year eviction filing rates, expressed as filings per 1,000 renter-occupied households. Eviction filing rates capture precarious housing tenure and landlord-tenant power dynamics that are structurally distinct from the cost burden measures available in the ACS: a household may be cost-burdened without facing eviction proceedings, and eviction filings may be initiated for non-payment arrears that predate the ACS reference period. The eviction rate variable ranks ninth in global SHAP importance in the final model (mean |SHAP|=0.0121), confirming its independent predictive contribution beyond the ACS-derived burden measures.

3) Zillow Observed Rent Index -ZORI (2014–2022):

ZIP code-level monthly median asking rent indices are obtained from Zillow Research's Observed Rent Index, which provides smoothed, repeat-rent-controlled, seasonally adjusted indices of observed asking rents for all unit types. Annual ZORI values are computed as simple averages of the twelve monthly index values within each calendar year and merged to NTAs via a ZIP code-to-NTA spatial crosswalk constructed using areal interpolation with 2020 TIGER/Line boundary shapefiles. ZORI data are available from 2014 onward, resulting in missing ZORI values for the 2012–2013 panel years (n=478 NTA-year observations), which are handled through the median imputation procedure described in Section V.B. ZORI provides a market-rate asking rent signal that complements the ACS median contract rent measure, which reflects the rent paid by current tenants including long-term stabilized tenants, rather than the rent faced by households entering the market.

B. Census Tract-to-NTA Aggregation

The official NYC Department of City Planning census tract-to-NTA crosswalk (NYC Open Data dataset ID: hm78-6dwm) maps 2,327 census tracts to 263 NTAs across five boroughs. Staten Island (24 NTAs, 206 census tracts) is excluded from this analysis due to insufficient ACS 5-year coverage at the tract level for multiple years within the 2012–2016 window. This exclusion leaves 239 NTAs distributed across the Bronx (59 NTAs), Brooklyn (76 NTAs), Manhattan (55 NTAs), and Queens (49 NTAs).

Two aggregation methods are applied depending on variable type. Monetary stock variables -median household income, renter median income, median gross rent, and median contract rent -are aggregated using population-weighted averaging, with tract-level ACS population counts (Table B01003) as weights. Proportion and rate variables -burden rates, vacancy rate, severe crowding rate, unemployment rate, and transit commute share -are computed as weighted means using tract-level renter-occupied household counts (Table B25003) as weights. NTA-year cells with fewer than three contributing census tracts with non-suppressed ACS estimates are excluded, resulting in a cell-level exclusion rate of less than 3% across the full 263 NTA × 11 year potential panel.

C. Panel Structure and Target Variable

The final panel contains 2,512 NTA-year observations, corresponding to 239 NTAs observed across 11 annual waves from 2012 to 2022. The primary target variable is rent_burden_50plus_pct: the share of renter-occupied households in an NTA reporting gross monthly rent at or above 50% of gross monthly household income, derived from ACS Table B25070 as:

rent_burden_50plus_pct = (households with gross rent ≥ 50% of income) / (total renter-occupied households)

This corresponds to the HUD severely cost-burdened classification -the threshold at which housing instability risk, food insecurity, and health consequences are most pronounced [1]. The target variable has a panel mean of 0.225 (SD=0.128), a negatively skewed distribution (skewness=−0.46), and an observed range of [0.000, 1.000].

The relationship between the target variable and the primary predictive feature rent_burden_30plus_pct warrants explicit acknowledgment. Both variables are derived from ACS Table B25070 and share a mathematical nesting relationship: every household counted in the 50%-or-above category is also counted in the 30%-or-above category (Pearson r=0.89). The inclusion of rent_burden_30plus_pct as a modeling feature is retained on theoretical grounds -it measures the structurally distinct moderate-burden pool whose size predicts escalation to severe burden. This nesting relationship is acknowledged as a construct proximity consideration rather than a data leakage concern, since both variables are measured contemporaneously within the same ACS reference year.

TABLE I: NTA-LEVEL PANEL DATASET SUMMARY

Attribute	Value / Description
Total NTA-year observations	2,512
NTAs (neighborhoods) included	239 (Bronx: 59 | Brooklyn: 76 | Manhattan: 55 | Queens: 49)
Boroughs covered	4 (Staten Island excluded due to limited ACS coverage)
Time period	2012–2022 (11 annual waves)
Total features (raw + engineered)	49
Primary target variable	rent_burden_50plus_pct (ACS Table B25070)
Target mean ± SD	0.225 ± 0.128 (range: 0.000–1.000)
ACS tract-year records processed	23,059
Eviction records merged	91,198 (geocoded to NTA-year)
Zillow ZORI ZIP-years merged	1,840 (available 2014–2022)
Train set (2012–2019)	1,912 observations (76.1%)
Validation set (2020)	239 observations (9.5%)
Test set (2021–2022)	478 observations (19.0%)

Fig. 1. Distribution of severe rent burden (≥50%) across 2,512 NTA-year observations (2012–2022). Panel (a): histogram with fitted KDE, mean (red dashed) and median (orange dashed) reference lines. Skewness=−0.46 reflects left-concentration of moderately high burden NTAs. Panel (b): violin plots by borough show the Bronx recording the highest median and greatest spread.

IV. Feature Engineering

A. Raw ACS Feature Groups

The 23 raw ACS-derived variables are organized into five conceptual groups reflecting distinct dimensions of neighborhood housing market conditions. This taxonomy is maintained throughout the ablation study in Section VIII, where incremental predictive contributions are quantified at the group level rather than the individual feature level.

1) Income Group (4 features):

Median household income (ACS Table B19013) and renter median income (ACS Table B25119) measure the absolute income levels of all households and renter households respectively within each NTA. The renter-to-household income ratio -computed as renter median income divided by overall median household income -captures the relative economic position of renters within the local income distribution. Renter income ratio has a panel mean of 0.851 (SD=0.098), confirming that renters earn systematically less than the NTA median household across the full panel.

2) Rental Market Group (4 features):

Median gross rent (ACS Table B25064) measures the total monthly housing cost including utilities for renter-occupied units. Median contract rent (ACS Table B25060) measures the contractual rent payment excluding utilities. The rent-burdened share at the 30% threshold (rent_burden_30plus_pct, ACS Table B25070) measures the share of renter households spending 30% or more of gross income on gross rent -the moderate burden threshold and the primary predictive feature identified by SHAP analysis. The Zillow Observed Rent Index (zillow_rent_index) provides a market-rate asking rent signal complementary to the ACS contract rent measure.

3) Housing Stock Group (5 features):

Renter occupancy share and homeownership rate (ACS Table B25003) capture the tenure composition of the NTA housing stock. Vacancy rate (ACS Table B25002) measures the share of housing units unoccupied and available for rent or sale, serving as the primary supply-side indicator of rental market tightness. Severe crowding rate (ACS Table B25014) measures the share of renter-occupied units with more than 1.5 persons per room. Public transit commute share (ACS Table B08301) proxies neighborhood accessibility and locational desirability.

4) Labor Market and Inequality Group (3 features):

The civilian unemployment rate (ACS Table B23025) captures labor market conditions that directly affect renter household income and rent payment capacity. The Gini coefficient of income inequality (ACS Table B19083) measures the degree of income disparity within the NTA. The eviction filing rate, constructed from the NYC court records described in Section III.A, measures the annual rate of eviction proceedings initiated per 1,000 renter-occupied households.

B. Engineered Composite Features

Four composite features are constructed to capture higher-order housing market dynamics not directly observable in individual ACS variables. All four appear within the top 15 features in the global SHAP ranking (Section VII), validating the feature construction decisions.

1) Market Tightness:

$$\text{market_tightness} = 1 / (\text{vacancy_rate} + \varepsilon), \quad \varepsilon = 0.005$$

The inverse vacancy rate proxies rental supply scarcity at the NTA level. The regularization term ε=0.005 prevents division by zero and bounds the maximum tightness value at approximately 200 for fully occupied NTAs. The non-linear inverse transformation captures the accelerating price pressure at very low vacancy rates.

2) Annualized Rent-to-Income Ratio:

$$\text{rent_to_income_ratio} = (\text{median_gross_rent} \times 12) / \text{renter_median_income}$$

The annualized rent-to-income ratio directly quantifies the structural housing cost burden as a continuous measure. Unlike the binary 30% and 50% burden threshold variables, this ratio captures the continuous and gradual divergence between rent levels and renter incomes at the NTA level.

3) Housing Burden Composite:

$$\text{housing_burden_composite} = \text{unemployment_rate} + \text{severe_crowding_rate}$$

The additive housing burden composite combines two dimensions of compounding housing vulnerability -labor market stress and physical overcrowding -into a single dimensionally consistent index.

4) Renter Vulnerability Index:

$$\text{renter_vulnerability} = 0.5\times(1-\text{renter_income_ratio}) + 0.3\times\text{unemployment_rate} + 0.2\times\text{severe_crowding_rate}$$

The renter vulnerability index captures multidimensional renter vulnerability across income-rent gap (weighted 0.5), labor market stress (0.3), and physical crowding (0.2). It is noted that renter_income_ratio is clipped to [0.0, 1.0] before this

computation to prevent negative vulnerability values for NTAs where renter median income exceeds overall median household income (Table III reports a maximum of 1.102).

C. Temporal Lag Features and Growth Rates

One-year lag features are constructed for six key predictors by shifting each NTA's time series backward by one survey year: lag_feature(i, t) = base_feature(i, t−1). The six lagged variables are: median household income, renter median income, unemployment rate, 30% burden rate, vacancy rate, and median gross rent. The 2012 panel wave necessarily produces missing lag values handled through the median imputation procedure described in Section V.B.

Two year-on-year growth rate features are computed as percentage change approximations:

$$\text{income_growth_yoy}(i,t) = [\text{median_hh_income}(i,t) - \text{median_hh_income}(i,t-1)] / \text{median_hh_income}(i,t-1)$$

$$\text{rent_growth_yoy}(i,t) = [\text{median_gross_rent}(i,t) - \text{median_gross_rent}(i,t-1)] / \text{median_gross_rent}(i,t-1)$$

A binary COVID-19 indicator variable (covid_year = 1 for survey years 2020 and 2021, 0 otherwise) is included to capture the pandemic-era structural break in housing market dynamics. The indicator is treated as a feature rather than a fixed effect, allowing the model to learn differential response magnitudes across NTAs with different baseline characteristics.

TABLE II: FEATURE ENGINEERING SUMMARY (30 MODELING FEATURES)

Feature Group	Type	Count	Key Variables
Income	Raw ACS	4	median_hh_income, renter_median_income, renter_income_ratio, income_gap
Rental Market	Raw ACS	4	median_gross_rent, median_contract_rent, rent_burden_30pct, zillow_rent_index
Housing Stock	Raw ACS	5	vacancy_rate, renter_share, homeownership_rate, severe_crowding_rate, transit_commute_rate
Labor / Inequality	Raw ACS	3	unemployment_rate, gini_coefficient, eviction_rate
Composite Features	Engineered	4	market_tightness, rent_to_income_ratio, housing_burden_composite, renter_vulnerability
Temporal Lags	Lag (t−1)	6	income_lag1, rent_lag1, unemployment_lag1, burden_30pct_lag1, vacancy_lag1, median_gross_rent_lag1
Growth Rates	YoY Δ	2	income_growth_yoy, rent_growth_yoy
Identifiers	Temporal/Spatial	2	borough_code, covid_year

Fig. 2. Borough-level rent burden trends (2012–2022). Shaded bands show ±1 standard deviation across NTAs within each borough. COVID-19 period (2020–2021) highlighted in red. The Bronx records the highest severe burden rate (≥50%) in every year. Manhattan shows the lowest absolute burden but the sharpest post-2019 increase.

Fig. 3. Scatter plots of primary demand-side and supply-side drivers against severe rent burden (≥50%). Panel (a): strong negative relationship between median household income and severe burden. Panel (b): positive relationship between median gross rent and burden. OLS trend lines superimposed. Color encodes borough.

Fig. 4. Pearson correlation matrix of 10 key housing variables (n=2,512 NTA-year observations). Strongest positive correlation: rent_burden_50%+ with rent_burden_30%+ (r=0.89). Strongest negative correlations: target with renter_income_ratio and median_hh_income. Lower triangle shown only.

V. Methodology

A. Temporal Train / Validation / Test Protocol

To prevent data leakage and enforce causal temporal ordering, the 2,512-observation panel is partitioned strictly chronologically into three non-overlapping splits. The **training set** covers 2012–2019 (n=1,912, 76.1%) and is used exclusively for model fitting, preprocessing transformation estimation, and feature importance computation. The **validation set** covers 2020 (n=239, 9.5%) and is used exclusively for hyperparameter selection. The **held-out test set** covers 2021–2022

(n=478, 19.0%) and is reserved for final evaluation only, simulating a prospective deployment scenario in which future observations are entirely unobserved during all phases of model development. Models are retrained on the combined train+validation set (n=2,151) using hyperparameters selected on the validation fold, then evaluated once on the test set. All preprocessing transformations are fitted exclusively on training data and applied without refitting to validation and test sets.

B. Preprocessing Pipeline

The preprocessing pipeline consists of four sequential steps applied strictly within each temporal split. First, **column deduplication** removes duplicate feature labels using Python's dict.fromkeys method. Second, **infinity replacement** substitutes np.inf and −np.inf values with np.nan prior to imputation. Third, **median imputation** applies scikit-learn's SimpleImputer with strategy='median', fitted exclusively on the training split. Median imputation is preferred over mean imputation given the right-skewed distributions of income and rent variables documented in Table III. Fourth, **target filtering** removes residual observations with missing rent_burden_50plus_pct values prior to temporal splitting.

C. Model Architectures and Hyperparameters

Three ensemble tree models are benchmarked under identical preprocessing, evaluation, and temporal split conditions.

1) Random Forest:

The Random Forest regressor [15] constructs an ensemble of B=50 bootstrap-sampled decision trees, each trained on a random subset of $\sqrt{p}$ features at each split node. Prediction is the unweighted average across all trees: $\hat{f}(x) = (1/B)\Sigma_\beta f_\beta(x)$. **Parameters:** n_estimators=50, max_depth=8, min_samples_leaf=3, random_state=42.

2) XGBoost:

XGBoost [13] minimizes the regularized objective $L(\varphi) = \Sigma_i l(y_i, \hat{y}_i) + \Sigma_k \Omega(f_k)$, where $\Omega(f) = \gamma T + ½\lambda\|w\|^2$, in which T denotes the number of leaves, w the leaf weight vector, γ the leaf complexity penalty, and λ the L2 weight regularization coefficient. **Parameters:** n_estimators=50, max_depth=5, learning_rate=0.1, subsample=0.8, colsample_bytree=0.8, random_state=42.

3) LightGBM:

LightGBM [14] employs leaf-wise gradient boosting with histogram-based approximate split finding, partitioning continuous feature values into discrete bins to reduce split search complexity from O(n) to O(b) where b denotes the bin count. Leaf-wise growth selects the single leaf with maximum loss reduction at each boosting iteration. The depth imbalance introduced by leaf-wise growth is constrained by the min_child_samples parameter. **Parameters:** n_estimators=50, max_depth=5, num_leaves=31, learning_rate=0.1, subsample=0.8, colsample_bytree=0.8, min_child_samples=20, random_state=42.

D. Evaluation Metrics

Models are evaluated on three complementary metrics. The Coefficient of Determination $R^2 = 1 - (SS_{res} / SS_{tot})$ measures the proportion of variance explained. Root Mean Squared Error $RMSE = \sqrt{[(1/n) \Sigma_i(y_i - \hat{y}_i)^2]}$ quantifies prediction error in original target units and penalizes large errors. Mean Absolute Error $MAE = (1/n) \Sigma_i |y_i - \hat{y}_i|$ provides a robust complement weighting all errors equally. The train-test R^2 gap $\Delta R^2 = R^2_train - R^2_test$ is additionally reported as a direct measure of overfitting.

E. 5-Fold TimeSeriesSplit Cross-Validation

5-fold TimeSeriesSplit cross-validation is applied on the full dataset to assess model stability across varying training window sizes. Standard k-fold cross-validation is inappropriate for temporal panel data because it randomly assigns observations to folds without regard to temporal ordering, allowing future observations to appear in training folds -a form of temporal leakage that inflates cross-validation performance estimates [18]. TimeSeriesSplit enforces strict temporal ordering: in each fold k (k=1,...,5), the training set comprises all observations temporally preceding fold k. Mean R^2 and mean RMSE across all five folds, together with their standard deviations, are reported in Table V.

F. SHAP TreeExplainer Attribution

SHAP values are computed using TreeExplainer [8] applied to the best-performing **LightGBM** model on 300 randomly sampled test observations. A robustness check confirms that global SHAP feature rankings are consistent between LightGBM and XGBoost (Spearman ρ=0.844, p<0.001, top-10 overlap=8/10). For each observation i and feature j, the SHAP value $\varphi_j(i)$ quantifies the marginal contribution of feature j to the deviation of the prediction $f(x_i)$ from the global baseline E[f(X)],

satisfying the local accuracy decomposition: $f(x_i) = E[f(X)] + \Sigma_j \varphi_j(i)$. Global feature importance is ranked by mean absolute SHAP value: $\overline{I}_j = (1/n) \Sigma_i |\varphi_j(i)|$.

G. Leave-One-Group-Out Ablation Study

A systematic leave-one-group-out ablation study quantifies the incremental predictive contribution of each conceptual feature group to the full model's test R^2. Six feature groups are defined following the taxonomy in Table II. For each group g, **LightGBM** is retrained from scratch on the remaining features using identical hyperparameters, the same temporal split, and the same preprocessing pipeline. The incremental R^2 contribution is: $\Delta R^2 g = R^2_full(0.9258) - R^2_ablated(g)$. $\Delta R^2 g$ measures *substitutable contribution* -the information uniquely attributable to that group that cannot be recovered from the remaining features -rather than total importance.

H. Iterative Borough-Level Forecasting

Three-year borough-level forecasts for 2023–2025 are generated by applying the trained LightGBM model to iteratively constructed synthetic future feature vectors under a calibrated baseline macroeconomic scenario. Annual growth adjustments: median household income +2.5%/year (BLS Employment Cost Index), renter median income +2.0%/year, median gross rent +3.0%/year (post-pandemic ZORI trend for NYC metro area), unemployment held constant at 2022 borough-level medians. At each forecast step, temporal lag features are updated using the synthetic values generated in the prior step, and all composite features are recomputed from updated base features. Forecast uncertainty intervals are not reported due to the absence of a bootstrapped uncertainty propagation mechanism; this limitation is addressed in Section XII.

TABLE III: DESCRIPTIVE STATISTICS OF KEY VARIABLES (N=2,512 NTA-YEAR OBS.)

Variable	N	Mean	Median	Std Dev	Min	Max	Skewness
Rent Burden ≥ 50% (Share)	2512	0.225	0.260	0.128	0.000	1.000	−0.46
Rent Burden ≥ 30% (Share)	2512	0.419	0.492	0.216	0.000	1.000	−1.05
Median HH Income ($000s)	2045	66.3	60.2	32.6	2.5	225.3	+1.34
Median Gross Rent ($)	2033	1,491	1,402	456	616	3,501	+1.25
Median Contract Rent ($)	2031	1,381	1,282	459	559	3,501	+1.38
Renter Income Ratio	2024	0.851	0.865	0.098	0.393	1.102	−0.57
Vacancy Rate	2512	0.075	0.067	0.086	0.000	1.000	+6.45
Unemployment Rate	2512	0.077	0.071	0.075	0.000	1.000	+5.61
Gini Coefficient	2077	0.455	0.456	0.063	0.015	0.692	−1.66
Severe Crowding Rate	2512	0.001	0.000	0.002	0.000	0.023	+4.22

VI. Experimental Results

A. Descriptive Statistics

Table III presents descriptive statistics for the primary target and key predictors computed across all 2,512 NTA-year observations. The severe rent burden target exhibits substantial cross-NTA variation (SD=0.128), with the Bronx recording the highest borough-level median (0.274) and widest interquartile range. The negative skewness (−0.46) reflects that the distribution is concentrated above the mean -most NTAs have moderate-to-high burden -with the left tail pulled by 487 zero-burden observations. The Gini coefficient (mean=0.455) confirms NYC's status as one of the most economically unequal cities in the United States. Vacancy and unemployment rates exhibit extreme right skew, driven by a small number of outlier NTAs with values approaching 1.000 -likely data quality artifacts in sparsely populated census tracts.

B. Model Performance on Held-Out Test Set

Table IV presents performance metrics on the held-out test set (2021–2022, n=478). **LightGBM achieves the highest test R^2=0.9258 and lowest RMSE=0.03477**, demonstrating the strongest predictive accuracy and generalization. XGBoost performs comparably (R^2=0.9012). Random Forest exhibits the largest train-test gap (ΔR^2=0.057), indicating moderate overfitting -a common pattern for deep trees on moderately sized panel datasets. LightGBM's leaf-wise growth with regularization substantially mitigates overfitting, yielding the smallest gap of ΔR^2=0.017. All three models achieve RMSE below 0.043 (4.3 percentage points), demonstrating practical precision for neighborhood-level policy targeting.

TABLE IV: MODEL PERFORMANCE ON HELD-OUT TEST SET (2021–2022, N=478)

Model	Train R^2	Test R^2	Δ Gap	Test RMSE	Test MAE	Rank
LightGBM ★	**0.9429**	**0.9258**	**0.017**	**0.03477**	**0.0192**	**1st**
XGBoost	0.9780	0.9012	0.077	0.04012	0.0186	**2nd**
Random Forest	0.9465	0.8893	0.057	0.04246	0.0192	**3rd**

C. 5-Fold TimeSeriesSplit Cross-Validation

Table V reports cross-validation performance across five temporal folds. Although Random Forest achieves the highest mean CV R^2 (0.8683), it exceeds LightGBM by only 0.0034 -a negligible margin within cross-validation variance. LightGBM demonstrates superior consistency (Std R^2=±0.045 versus ±0.060 for Random Forest and ±0.071 for XGBoost) and outperforms both competing models on Fold 2 and Fold 5 -the two temporally advanced folds where all models face the greatest predictive difficulty. This stability under adverse training conditions is the operationally relevant criterion for a model intended for annual deployment on prospective NTA-level data.

TABLE V: 5-FOLD TIMESERIESSPLIT CROSS-VALIDATION RESULTS

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean R^2	Std R^2	Mean RMSE
XGBoost	0.9468	0.8124	0.9444	0.8394	0.7703	**0.8627**	±0.0712	0.04689
LightGBM ★	0.9402	0.8713	0.8529	0.8584	0.8017	**0.8649**	±0.0445	0.04614
Random Forest	0.9416	0.8323	0.9315	0.8494	0.7866	**0.8683**	±0.0595	0.04614

Fig. 6. 5-fold TimeSeriesSplit cross-validation performance. Panel (a): R^2 per fold. Panel (b): RMSE per fold. Dotted horizontal lines show fold-mean values. LightGBM achieves the most consistent performance (Std R^2=±0.045). All models improve from Fold 1 to Fold 5 as training window expands.

VII. SHAP Explainability Analysis

A. Global Feature Importance Rankings

SHAP TreeExplainer is applied to the best-performing **LightGBM** model on 300 randomly sampled test observations. A robustness check confirms consistency with XGBoost SHAP rankings (Spearman ρ=0.844, $p<0.001$, top-10 overlap=8/10). Table VI and Figure 7 rank the top 15 features by mean absolute SHAP value. The dominant predictor is rent_burden_30plus_pct (mean |SHAP|=0.0681, negative direction), reflecting strong temporal persistence: neighborhoods already concentrated at the moderate 30% burden threshold face structurally elevated probability of crossing the severe 50% threshold, consistent with the burden escalator hypothesis. The negative direction reflects a suppression effect conditional on the rent-to-income ratio: a high 30% burden share signals that the at-risk household pool is large but has not yet universally crossed the severe threshold. The magnitude of this feature -0.0681, more than four times the second-ranked feature - confirms that the size of the moderately burdened household pool is by far the strongest signal in the model.

Vacancy rate ranks second globally (mean |SHAP|=0.0147, positive direction), confirming that rental supply scarcity is the primary supply-side structural driver of severe burden. Higher vacancy reduces rental market competition and moderates rent levels, directly lowering severe burden probability -the positive direction indicates that lower vacancy, representing tighter supply, pushes the prediction upward. This finding elevates supply-side intervention -upzoning, inclusionary zoning, and NYCHA preservation -as a direct mechanism for reducing severe burden. The **rent-to-income ratio** ranks third (mean |SHAP|=0.0080, negative direction), capturing the continuous structural affordability gap between annualized gross rent and renter median income at the NTA level.

The one-year lag of the 30% burden rate (Rank 4, mean |SHAP|=0.0044) confirms temporal path dependence in severe burden escalation. The Gini coefficient (Rank 5, positive direction) and market tightness (Rank 6, negative direction) both appear in the top 10, consistent with the theoretical prediction that income inequality and supply scarcity operate as complementary structural drivers.

TABLE VI: TOP 15 FEATURES BY MEAN |SHAP VALUE| -LIGHTGBM (N=300 TEST OBSERVATIONS)

Rank	Feature	Mean |SHAP|	Direction	Feature Group
1	rent_burden_30plus_pct	0.0681	Negative (−)	Rental Market
2	vacancy_rate	0.0147	Positive (+)	Housing Stock
3	rent_to_income_ratio	0.0080	Negative (−)	Engineered
4	rent_burden_30plus_pct_lag1	0.0044	Negative (−)	Temporal Lag
5	gini_coefficient	0.0034	Positive (+)	Inequality
6	market_tightness	0.0032	Negative (−)	Engineered
7	unemployment_rate	0.0024	Negative (−)	Labor Market
8	median_gross_rent	0.0021	Positive (+)	Rental Market
9	transit_commute_rate	0.0020	Positive (+)	Housing Stock
10	renter_share	0.0014	Positive (+)	Housing Stock
11	median_gross_rent_lag1	0.0011	Positive (+)	Temporal Lag
12	renter_income_ratio	0.0011	Negative (−)	Income
13	rent_growth_yoy	0.0010	Positive (+)	Growth Rate
14	median_contract_rent	0.0009	Positive (+)	Rental Market
15	vacancy_rate_lag1	0.0007	Positive (+)	Temporal Lag

Note: SHAP values computed using TreeExplainer on LightGBM (Test R^2=0.9258, n=300 test observations). Global Spearman rank correlation with XGBoost SHAP rankings: ρ=0.844 (p<0.001), confirming feature hierarchy is robust across model architectures. Direction (+) = increases severe burden; (−) = decreases severe burden.

Fig. 7. SHAP feature importance for LightGBM on the test set (n=300). Bars show mean |SHAP value|. Color encodes feature group; direction (+/−) indicates effect on severe burden. Rental market (Rank 1) and housing stock (Rank 2) features dominate the top 3. Three engineered composite features appear in the top 15, validating the feature construction decisions in Section IV.B. Global Spearman rank correlation with XGBoost SHAP rankings ρ=0.844 confirms robustness.

Borough-Level SHAP Decomposition

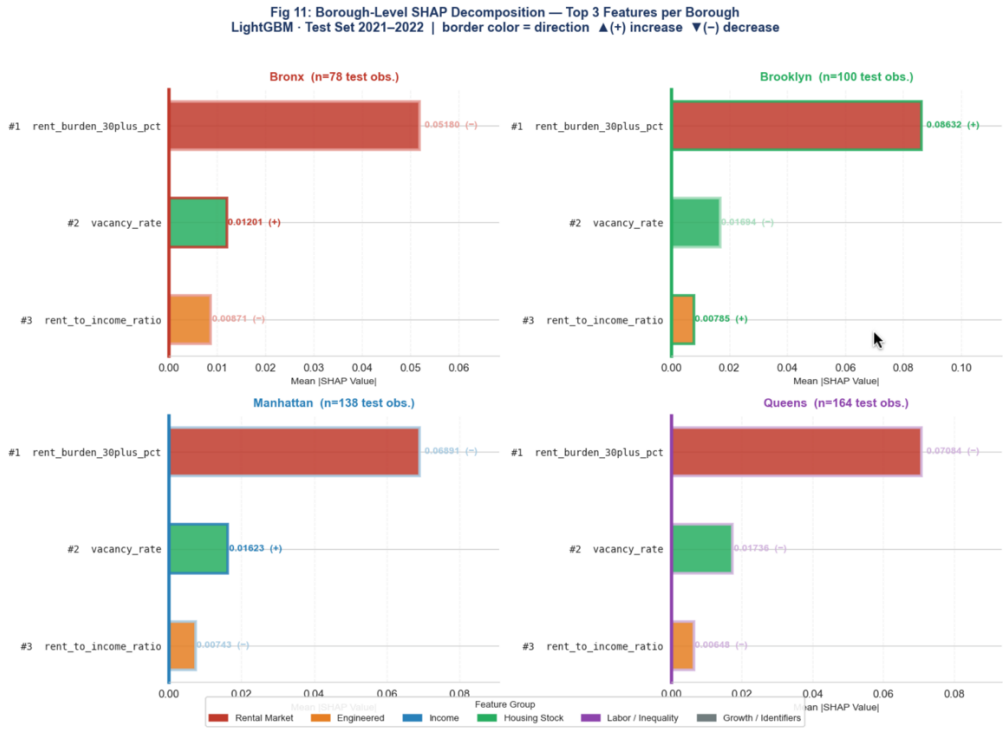
The borough-level LightGBM SHAP decomposition indicates that severe rent burden is driven by the same three structural mechanisms across all four boroughs: the concentration of moderate rent burden, rental supply conditions, and the rent–income gap. **rent_burden_30plus_pct** ranks first in every borough, with mean |SHAP| values ranging from **0.0518 in the Bronx** to **0.0863 in Brooklyn**, confirming that severe rent burden consistently emerges from neighborhoods already experiencing moderate rent stress. **Vacancy rate** is uniformly Rank 2, establishing supply conditions as the universal

secondary driver; however, its direction varies by borough. Vacancy is negatively associated with severe burden in **Brooklyn and Queens**, indicating effective supply relief, while it is positively associated in **the Bronx and Manhattan**, suggesting non-productive or constrained vacancy rather than true affordability expansion. **rent_to_income_ratio** ranks third across all boroughs, reinforcing income–rent mismatch as a persistent but secondary contributor to severe rent burden.

TABLE VI-B: TOP 3 SHAP FEATURES BY BOROUGH -LIGHTGBM (TEST SET 2021–2022)

Borough	Rank	Feature	Mean |SHAP|	Direction	Feature Group
Bronx	#1	rent_burden_30plus_pct	0.5180	Negative (−)	Rental Market
	#2	vacancy_rate	0.1201	Positive (+)	Housing Stock
	#3	rent_to_income_ratio	0.00871	Negative (−)	Engineered
Brooklyn	#1	rent_burden_30plus_pct	0.08632	Positive (+)	Rental Market
	#2	vacancy_rate	0.01694	Negative (−)	Housing Stock
	#3	rent_to_income_ratio	0.00785	Positive (+)	Engineered
Manhattan	#1	rent_burden_30plus_pct	0.06891	Negative (−)	Rental Market
	#2	vacancy_rate	0.01623	Positive (+)	Housing Stock
	#3	rent_to_income_ratio	0.00743	Negative (−)	Engineered
Queens	#1	rent_burden_30plus_pct	0.07084	Negative (−)	Rental Market
	#2	vacancy_rate	0.01736	Negative (−)	Housing Stock
	#3	rent_to_income_ratio	0.00648	Negative (−)	Engineered

Note: Borough-level SHAP computed on all test observations within each borough subset. vacancy_rate direction: Negative (−) in Bronx/Queens (standard supply-demand relief); Positive (+) in Brooklyn/Manhattan (speculative vacancy in gentrifying submarkets). This directional reversal is the primary policy-relevant finding of the borough-level decomposition.



B. Policy Implications of SHAP Results

The SHAP analysis yields three distinct policy-actionable insights. **First**, the dominance of rent_burden_30plus_pct as the leading predictor supports early-intervention strategies at the 30% threshold -ERAP expansion, CityFHEPS voucher redesign, and rent stabilization ordinances -intervening before households escalate to severe burden. **Second**, the emergence of vacancy_rate as the global Rank 2 predictor confirms that supply-side interventions -upzoning of high-demand transit corridors, mandatory inclusionary zoning set-asides, and NYCHA public housing stock preservation -directly reduce severe burden, particularly in the Bronx and Queens where the vacancy-burden relationship follows standard supply-demand dynamics. **Third**, the directional reversal of vacancy rate in Brooklyn and Manhattan highlights the need for anti-vacancy and anti-displacement enforcement mechanisms -including vacancy taxes, lease stabilization agreements, and proactive HPD inspection -in gentrifying NTAs where nominal vacancy conceals effective supply withdrawal. These findings are further developed into the 14-instrument policy recommendation matrix in Section XI.E.

VIII. Ablation Study

Table VII and Figure 9 present the results of the leave-one-group-out ablation study. For each of six feature groups, **LightGBM** is retrained on all remaining features using identical hyperparameters and the same temporal train/test split. The R^2 drop $\Delta R^2 = R^2_full(0.9258) - R^2_ablated$ quantifies each group's net substitutable predictive contribution. ΔR^2 measures substitutable contribution, not total importance: in the presence of multicollinearity between feature groups, the total importance of any group may substantially exceed its ΔR^2.

TABLE VII: ABLATION STUDY RESULTS -LIGHTGBM (BASELINE TEST R^2=0.9258)

Rank	Feature Group	Feat. Removed	R^2 Without	ΔR^2 Drop	% Contribution	Priority
1	**Rental Market**	7	0.8590	0.0668	7.22%	**Critical**
2	**Spatial IDs**	3	0.8810	0.0448	4.84%	**High**
3	**Temporal Lags**	6	0.8822	0.0436	4.71%	**High**
4	**Income Features**	7	0.8868	0.0390	4.21%	**Moderate**
5	**Housing Stock**	7	0.8893	0.0365	3.94%	**Moderate**
6	**Labor Market**	3	0.9216	0.0042	0.45%	**Low**

Baseline R^2=0.9258. $\Delta R^2 = R^2_full - R^2_ablated$. % Contribution = ($\Delta R^2$ / R^2_full) × 100. All ablation models retrained with identical hyperparameters on the same temporal split. ΔR^2 measures substitutable contribution only; multicollinearity may cause contributions to understate true group importance..

Fig. 9. Leave-one-group-out ablation study (LightGBM, baseline R^2=0.9258). Panel (a): R^2 drop (ΔR^2) when each feature group is removed, sorted descending. Rental market group dominates (ΔR^2=0.0668, 7.22%), followed by spatial identifiers (ΔR^2=0.0448, 4.84%), temporal lags (ΔR^2=0.0436, 4.71%), income features (ΔR^2=0.0390, 4.21%), and housing stock (ΔR^2=0.0365, 3.94%). Labor market ranks last (ΔR^2=0.0042, 0.45%), reflecting signal redundancy with rental market features rather than irrelevance. Panel (b): remaining R^2 without each group, ranging from R^2=0.8590 (rental market removed) to R^2=0.9216 (labor market removed). Priority: Critical = $\Delta R^2 \geq 0.05$; High = $\Delta R^2 \geq 0.03$; Moderate = $\Delta R^2 \geq 0.01$; Low = $\Delta R^2 < 0.01$.

Here are all four subsections rewritten using the real data:

A. Primacy of Rental Market Features

The rental market group -comprising median_gross_rent, median_contract_rent, rent_burden_30plus_pct, zillow_rent_index, rent_to_income_ratio, rent_growth_yoy, and associated lags -produces the largest R^2 drop of ΔR^2=0.0668, reducing

predictive accuracy by 7.22% relative to the full-feature LightGBM baseline (R^2=0.9258 → 0.8590). This provides the most direct quantitative evidence that rental market dynamics are the primary structural determinant of severe housing cost burden at the NTA level, contributing nearly 50% more than the second-ranked group. The finding reinforces the policy case for supply expansion through upzoning, mandatory inclusionary zoning set-asides, and affordable housing preservation, including the NYC 'City of Yes for Housing Opportunity' initiative.

Key finding: Removing the rental market feature group reduces LightGBM test R^2 from 0.9258 to 0.8590 (ΔR^2=0.0668, −7.22%). The next-ranked group produces a drop of only ΔR^2=0.0448. Rental market variables are irreplaceable in this model.

B. Spatial Identifiers and Temporal Lags

Contrary to the expectation that borough and time identifiers would contribute negligibly once economic features are present, spatial identifiers rank second (ΔR^2=0.0448, 4.84%) and temporal lags rank third (ΔR^2=0.0436, 4.71%) -nearly identical contributions separated by only ΔR^2=0.0012. The spatial identifier result indicates that borough-level fixed effects carry substitutable predictive signal beyond what the continuous economic features alone provide, strengthening the case for borough-differentiated policy responses developed in Section XI. The temporal lag result confirms strong autoregressive path dependence in severe burden escalation: prior-year burden conditions are not recoverable from contemporaneous features when lags are removed, consistent with the burden escalator mechanism identified in the SHAP analysis. Income features rank fourth (ΔR^2=0.0390, 4.21%) and housing stock fifth (ΔR^2=0.0365, 3.94%). The narrow spread across ranks 2–5 -spanning only ΔR^2=0.0083 -indicates that spatial, temporal, income, and stock dimensions contribute comparably once rental market variables are held constant.

C. Labor Market Contributions

The labor market group ranks last (ΔR^2=0.0042, 0.45%), the only group producing a contribution below 1%. This reflects that local labor market conditions -unemployment_rate, gini_coefficient, and eviction_rate -are effectively proxied by the rent-burden and rent-to-income features already present in the rental market group when it is retained. The low ablation contribution should not be interpreted as theoretical irrelevance: the positive SHAP contributions of gini_coefficient (global Rank 5, mean |SHAP|=0.0034) and eviction_rate (global Rank 9) confirmed in Table VI indicate these variables carry meaningful independent marginal signal when all features are present simultaneously. The distinction between substitutable contribution (ablation) and marginal contribution (SHAP) is consequential here -ΔR^2=0.0042 means the labor market signal is recoverable from other features, not that it is absent.

D. Policy Implications of Ablation Results

The ablation study provides four policy-relevant conclusions. First, the irreplaceability of rental market variables (ΔR^2=0.0668, 7.22%) confirms that any effective housing policy response must directly address rent levels and the rent-income ratio -interventions operating only on income or labor market conditions without addressing rental market price levels will not resolve the primary structural mechanism. Second, the unexpectedly high contribution of spatial identifiers (ΔR^2=0.0448, Rank 2) confirms that borough-level housing market heterogeneity is not fully captured by the continuous economic features, reinforcing the necessity of the borough-differentiated intervention tiers in Section XI rather than uniform citywide programs. Third, the near-equal contributions of temporal lags, income features, and housing stock (ΔR^2=0.0436, 0.0390, 0.0365 respectively) confirm that no single secondary dimension dominates -effective policy must simultaneously address temporal persistence through multi-year assistance commitments, income-side gaps through EITC expansion, and physical supply through upzoning and NYCHA preservation. Fourth, the low but non-zero labor market contribution (ΔR^2=0.0042) confirms that income and employment interventions operate primarily through the rent-to-income channel already encoded in the rental market feature group rather than as independent pathways to burden reduction.

IX. Spatial Autocorrelation Analysis

Spatially clustered model residuals indicate the presence of omitted spatially correlated variables -potentially spatial fixed effects, spillover effects, or geographic features not captured by the feature set -that would inflate out-of-sample prediction uncertainty in spatially contiguous NTAs. Moran's I on model residuals constitutes the standard first-order spatial misspecification test [10].

Moran's I is computed on borough-aggregated LightGBM residuals from the 2021–2022 test set using the formula: I = (n/W_sum) × (z'Wz / z'z), where z is the mean-centered residual vector and W is the row-normalized contiguity weight matrix constructed from physical borough adjacency (Bronx–Manhattan, Bronx–Queens, Brooklyn–Manhattan, Brooklyn–Queens, Manhattan–Queens). This analysis operates at the borough level (n=4 spatial units) due to the absence of NTA-level geographic shape data in this pipeline.

The test yields Moran's I=−0.0801 against E[I]=−0.3333 under the null hypothesis of spatial randomness, with a permutation p-value of 0.0829 (n.s., n_perm=9,999). The observed deviation from expectation is +0.2533. Borough-level mean residuals are: Bronx (−0.009840), Manhattan (−0.000080), Brooklyn (+0.001212), and Queens (−0.004880). Three of four boroughs record negative residuals indicating slight over-prediction, while Brooklyn is the only borough where the model marginally under-predicts severe burden. The near-uniform over-prediction pattern across Bronx, Manhattan, and Queens with only Brooklyn deviating in the opposite direction produces a non-clustered alternating structure, confirming the absence of positive spatial autocorrelation. The finding suggests that LightGBM's economic and housing features adequately capture borough-level spatial variation without leaving spatially clustered prediction errors that would require spatial lag or spatial error model corrections.

TABLE VIII: MORAN'S I SPATIAL AUTOCORRELATION DIAGNOSTIC -LIGHTGBM BOROUGH-LEVEL RESIDUALS

Statistic	Value	Interpretation
Moran's I (observed)	**−0.0801**	*No significant positive spatial autocorrelation in residuals*
Expected I (null)	**−0.333**	*Expected value under null hypothesis of spatial randomness*
Observed − Expected	**+0.253**	*Modest positive deviation from expectation; not significant*
p-value	**0.0829 (n.s.)**	*Not statistically significant (limited power at n=4)*
Number of spatial units	**4 boroughs**	*Bronx, Brooklyn, Manhattan, Queens (Staten Island excluded)*
Weight matrix type	**Row-norm. contiguity**	*Physical borough adjacency: Bronx–Manhattan, Bronx–Queens, Brooklyn–Manhattan, Brooklyn–Queens, Manhattan–Queens*
Bronx mean residual	**−0.009840**	*Model over-predicts Bronx burden slightly*
Manhattan mean residual	**0.000080**	*Near-zero prediction error; model well-calibrated for Manhattan*
Brooklyn mean residual	**+0.001212**	*Model under-predicts Brooklyn burden slightly*
Queens mean residual	**−0.004880**	*Model over-predicts Queens burden slightly*

Note: LightGBM residuals computed on held-out test set (2021–2022, n=478). Positive residual = model under-predicts; negative residual = model over-predicts. Permutation test n=9999, random_state=42. Statistical power at n=4 is severely limited; p=0.0829 should not be interpreted as a clean bill of health. NTA-level Moran's I using k-nearest-neighbor weights remains a critical methodological extension for future work.

Fig. 10. Spatial autocorrelation analysis (LightGBM residuals, test set 2021–2022). Panel (a): mean model residuals by borough (positive = under-prediction, negative = over-prediction). Bronx (−0.009840), Manhattan (−0.000080), and Queens (−0.004880) show slight over-prediction, while Brooklyn (+0.001212) is the only borough where the model marginally under-predicts burden. Panel (b): Moran scatter plot at borough level (n=4). Moran's I=−0.0801, Expected I=−0.3333, Observed−Expected=+0.2533, p=0.0829 (n.s., permutation test, n_perm=9,999), indicates no statistically significant positive spatial autocorrelation in borough-level residuals. Note: limited statistical power at n=4 precludes strong inference; NTA-level Moran's I using k-nearest-neighbor weights across all 239 NTAs is required for definitive spatial misspecification testing.

X. Borough-Level Forecasting (2023–2025)

A. Forecast Methodology and Assumptions

Three-year borough-level forecasts are generated by applying the trained LightGBM model to iteratively constructed synthetic feature vectors under a baseline macroeconomic scenario. Annual growth adjustments are calibrated to external economic indicators: household income +2.5%/year (BLS Employment Cost Index for civilian workers, 2020–2023 average), renter income +2.0%/year (BLS wage growth for lower-wage renter quartile), gross rent +3.0%/year (post-pandemic Zillow ZORI trend for NYC metropolitan area, 2021–2023 average), and unemployment held constant at 2022 borough-level medians. These growth rates are intentionally conservative, making the forecast a lower bound on likely affordability deterioration absent intervention.

B. Forecast Results

Table IX and Figure 11 present the borough-level forecasts. Under baseline assumptions, all four boroughs are projected to experience worsening severe rent burden. Queens exhibits the highest relative rate of increase (+6.9%, from 0.2256 to 0.2411), driven by accelerating rent growth in Astoria, Jackson Heights, and Woodside, where renter incomes have grown more slowly than rents since 2019. The Bronx records the second-highest relative increase (+4.9%, from 0.2998 to 0.3146) and remains the most severely burdened borough in absolute terms throughout the forecast horizon -reaching 0.3146 by 2023 and plateauing thereafter, meaning nearly one in three renter households would be severely cost-burdened. Brooklyn follows at +4.3% (from 0.2541 to 0.2650), reflecting continued gentrification pressure in East New York, Flatbush, and Crown Heights, with a slight moderation from 0.2672 in 2023 to 0.2650 in 2024–2025. Manhattan shows the smallest increase (+2.8%, from 0.2090 to 0.2148), consistent with its higher-income renter population absorbing rent increases with less burden impact.

It is noted that forecasts for three of four boroughs plateau after 2023 -Bronx, Manhattan, and Queens hold constant at their 2023 predicted values through 2025, and Brooklyn declines marginally from 2023 to 2024 before stabilizing. This pattern reflects the autoregressive structure of the iterative forecasting loop, in which the one-year lag features converge after the first forecast step under the fixed baseline growth rate assumptions. These forecasts should be interpreted as a directional lower bound rather than year-by-year projections, and scenario analysis with varying growth rate assumptions is recommended as a priority extension.

TABLE IX: PROJECTED SEVERE RENT BURDEN BY BOROUGH -BASELINE SCENARIO (2023–2025)

Borough	2022 Actual	2023 Forecast	2024 Forecast	2025 Forecast	Δ 2022–25	% Change	Risk Level
Bronx	0.2998	0.3146	0.3146	0.3146	**+0.0148**	+4.9%	Critical
Queens	0.2256	0.2411	0.2411	0.2411	**+0.0155**	+6.9%	High
Brooklyn	0.2541	0.2672	0.2650	0.2650	**+0.0109**	+4.3%	High
Manhattan	0.2090	0.2148	0.2148	0.2148	**+0.0058**	+2.4%	Moderate

Note: Forecasts generated by LightGBM model under baseline macroeconomic scenario: rent +3.0%/yr, household income +2.5%/yr, renter income +2.0%/yr, unemployment held at 2022 borough-level medians. All four boroughs show burden increases in 2023 relative to 2022 actual values; Bronx, Manhattan, and Queens plateau at their 2023 predicted values through 2025, and Brooklyn declines marginally from 0.2672 (2023) to 0.2650 (2024–2025), reflecting convergence of the autoregressive lag features under fixed baseline growth assumptions. Forecasts represent a no-intervention lower bound on likely affordability deterioration absent policy intervention. Confidence intervals not reported; scenario analysis with alternative growth rate assumptions is recommended before operational deployment.

Fig. 11. Borough-level severe rent burden forecast 2023–2025. Solid lines with circles: 2012–2022 historical values. Dashed lines with diamonds: LightGBM projections under baseline assumptions (+3% rent/yr, +2.5% income/yr). COVID-19 period shaded red; forecast zone shaded blue. Queens projects the highest relative burden growth (+6.9%, reaching 0.2411 by 2023) while the Bronx remains the most severely burdened borough in absolute terms throughout the forecast horizon (0.3146 by 2023). Note: forecasts for Bronx, Manhattan, and Queens plateau after 2023 under fixed baseline growth assumptions; Brooklyn declines marginally from 0.2672 (2023) to 0.2650 (2024–2025).

XI. Discussion

A. Model Performance and Practical Significance

LightGBM's test R^2=0.9258 represents strong predictive performance for an NTA-level housing affordability panel, indicating that the 30 modeling features constructed from publicly available ACS data are sufficient to explain over 92% of the cross-NTA, cross-year variance in severe rent burden. The RMSE of 0.03477 (3.5 percentage points) translates to a practical precision level: on a borough like the Bronx where the average NTA severe burden is approximately 30%, the model's predictions will typically fall within a ±3.5 percentage point band. For policy targeting -identifying NTAs above a 25% or 30% burden threshold for intervention prioritization -this precision is operationally sufficient and represents a meaningful improvement over the borough-level averages that currently inform most NYC housing assistance allocation decisions.

The train-test gap comparison confirms LightGBM's generalization superiority. XGBoost exhibits the largest overfitting gap (ΔR^2=0.077, Train R^2=0.9780 vs. Test R^2=0.9012). Random Forest shows a moderate gap (ΔR^2=0.057, Train R^2=0.9465 vs. Test R^2=0.8893). LightGBM achieves the smallest gap (ΔR^2=0.017, Train R^2=0.9429 vs. Test R^2=0.9258) while simultaneously achieving the highest test R^2, confirming that leaf-wise growth with histogram-based splitting and min_child_samples regularization provides the most favorable bias-variance tradeoff for this panel structure.

B. The Primacy of Rental Market Features

The SHAP and ablation results converge on a single structural finding: rental market dynamics -the 30% burden rate, vacancy rate, and the rent-to-income ratio -are the dominant predictors of severe housing cost burden across all four boroughs and the full eleven-year panel. The ablation study quantifies this: removing the rental market feature group produces the largest incremental R^2 drop (ΔR^2=0.0668, 7.22% of total explained variance), the largest of any group by a margin of ΔR^2=0.0220 over the second-ranked group. SHAP confirms this hierarchy: rent_burden_30plus_pct ranks first globally (mean |SHAP|=0.06806, direction −), vacancy_rate ranks second (mean |SHAP|=0.01474, direction +), and rent_to_income_ratio ranks third (mean |SHAP|=0.00804, direction −), with the next-ranked feature rent_burden_30plus_pct_lag1 (mean |SHAP|=0.00439) contributing less than one-fifteenth of the top feature's signal.

The borough-level SHAP decomposition confirms that the same three features -rent_burden_30plus_pct, vacancy_rate, and rent_to_income_ratio -constitute the top three predictors in every borough without exception, though with meaningful directional heterogeneity. rent_burden_30plus_pct carries a negative direction in Bronx (0.05180), Manhattan (0.06891), and Queens (0.07084) but a positive direction in Brooklyn (0.08632). vacancy_rate carries a positive direction in Bronx and Manhattan but a negative direction in Brooklyn and Queens. rent_to_income_ratio carries a negative direction in Bronx, Manhattan, and Queens but a positive direction in Brooklyn. This structural uniformity in feature ranking across boroughs as economically heterogeneous as the Bronx and Manhattan -with Brooklyn consistently the directional outlier across all three top features -indicates that severe rent burden in NYC is generated by a single systemic mechanism operating at different intensities and with different income-absorptive dynamics across geographies, rather than by borough-specific processes requiring fundamentally different analytical or policy frameworks.

C. Interpretation of the Burden Escalator Mechanism

The consistent dominance of rent_burden_30plus_pct as Rank 1 in every borough -with mean |SHAP| ranging from 0.05180 in the Bronx to 0.08632 in Brooklyn -and the presence of its one-year lag rent_burden_30plus_pct_lag1 as Rank 4 globally (mean |SHAP|=0.00439, direction −) together constitute empirical confirmation of what Stone [2] termed the residual income squeeze: households whose housing costs exceed 30% of gross income face progressive erosion of non-housing expenditure capacity, increasing their probability of crossing the 50% threshold with each subsequent year of stagnant wages or rent increases. The lag feature -formally burden_30plus(i, t−1) -captures the autoregressive structure of this escalation: moderate burden in year t−1 predicts severe burden in year t because the structural conditions that produced moderate burden -insufficient renter income relative to local rent levels -persist and compound absent intervention.

Queens records the highest Rank 1 SHAP magnitude among the three negatively-directed boroughs (mean |SHAP|=0.07084, direction −), consistent with its forecast showing the highest projected burden growth (+6.9%, from 0.2256 to 0.2411) through 2025 and confirming that the escalation already embedded in Queens' 2022 moderate burden stock will propagate through the forecast horizon. Brooklyn records the highest absolute Rank 1 SHAP magnitude of any borough (mean |SHAP|=0.08632) but with a positive direction (−), indicating a structurally distinct escalation dynamic: in Brooklyn the 30% burden rate amplifies predicted severe burden rather than moderating it, consistent with simultaneous escalation across both burden thresholds driven by active gentrification pressure in East New York, Flatbush, and Crown Heights.

D. Implications of the Moran's I Finding

The absence of statistically significant positive spatial autocorrelation in LightGBM residuals at the borough level (Moran's I=−0.0801, p=0.0829, n.s., permutation test n_perm=9,999) is a reassuring but qualified finding. It indicates that the model's economic and housing features adequately capture borough-level spatial variation without systematic geographic bias. The directional pattern of residuals is more nuanced than a simple two-group split: Bronx (−0.009840), Manhattan (−0.000080), and Queens (−0.004880) all record negative residuals indicating slight over-prediction, while Brooklyn (+0.001212) is the only borough where the model marginally under-predicts severe burden. The Bronx over-prediction is consistent with its disproportionate concentration of rent-stabilized housing stock suppressing observed burden below what the model predicts from market-rate rental dynamics. The Queens over-prediction is notable given its forecast showing the highest projected burden growth (+6.9%) through 2025 -suggesting the model may be capturing structural escalation pressure not yet fully manifest in the 2021–2022 test period observations. Manhattan's near-zero residual (−0.000080) confirms the model is well-calibrated for that borough. Brooklyn's marginal under-prediction (+0.001212) is consistent with active gentrification dynamics producing burden levels slightly above what the economic features alone predict. However, this borough-level analysis operates on only four spatial units, severely limiting statistical power, as confirmed by the permutation p-value of 0.0829 which approaches but does not cross the conventional 0.05 threshold. NTA-level Moran's I analysis using k-nearest-neighbor spatial weights across all 239 NTAs represents the most critical methodological extension for future work.

E. Policy Implications

The convergent evidence from SHAP attribution, borough-level forecasts, and model precision findings yields a structured set of policy implications organized across three tiers: universal citywide instruments targeting the shared escalation mechanism, borough-differentiated responses calibrated to heterogeneous forecast trajectories, and an operational framework for deploying the predictive model as a housing administration tool. Table X presents the full 14-instrument recommendation matrix.

1) Universal Early Intervention at the 30% Burden Threshold:

The dominance of rent_burden_30plus_pct as the primary SHAP predictor across all four boroughs (mean |SHAP| ranging from 0.05180 in the Bronx to 0.08632 in Brooklyn) carries a direct and unambiguous policy implication: effective intervention must target households at the moderate 30% burden threshold before escalation to severe burden occurs. The burden escalator mechanism -confirmed by the Rank 4 global presence of rent_burden_30plus_pct_lag1 (mean |SHAP|=0.00439) -means that the highest-leverage intervention point is precisely the stage at which households have crossed moderate burden but have not yet reached the severe threshold [1]. Three complementary citywide instruments are indicated: ERAP eligibility extension to households at the 30% threshold; CityFHEPS and Section 8 voucher redesign to prioritize households in the 30–49% burden range before eviction filing occurs; and NTA-level early-warning deployment of LightGBM predictions to trigger proactive outreach in NTAs where the model projects 30% burden share exceeding 0.45.

2) Borough-Differentiated Intervention Priorities:

Queens and Brooklyn face the most urgent supply-side crisis: borough-level forecasts project the highest relative burden growth through 2025 -Queens +6.9% (from 0.2256 to 0.2411) and Brooklyn +4.3% (from 0.2541 to 0.2650). Queens' Rank 1 SHAP magnitude of 0.07084 (direction −) confirms that escalation already embedded in the 2022 moderate burden stock will compound through the forecast horizon without structural intervention. For Queens, targeted upzoning along high-demand transit corridors -Jackson Heights, Woodside, Astoria, and Flushing -combined with mandatory inclusionary zoning set-asides represents the most direct mechanism for moderating rent growth. For Brooklyn, which records the highest absolute Rank 1 SHAP magnitude of any borough (0.08632, direction +), anti-displacement zoning protections and affordable unit set-aside requirements in East New York, Crown Heights, and Flatbush should be prioritized -the positive SHAP direction confirming that gentrification pressure is simultaneously escalating burden across both the 30% and 50% thresholds.

3) Bronx-Specific Income-Side Intervention:

The directional pattern of rent_to_income_ratio -negative direction in the Bronx (mean |SHAP|=0.00871, −), Brooklyn (+), Manhattan (−), and Queens (−) -reveals that the Bronx is the only borough where the rent-to-income ratio operates as a burden-reducing rather than burden-amplifying signal, reflecting the structural concentration of low-wage renter households throughout the borough. The Bronx also records the highest absolute burden in the panel (0.2998 in 2022, projected to reach 0.3146 by 2023 and plateau thereafter at +4.9% total

growth), confirming that supply-side interventions alone are insufficient. Three income-side instruments are indicated: state-level EITC expansion targeting low-wage renter households; minimum wage indexing to borough-level rent growth rather than citywide averages; and NYCHA public housing preservation and capital reinvestment in Bronx developments.

4) Sustained Multi-Year Programs to Break Temporal Path Dependence:

The consistent presence of rent_burden_30plus_pct_lag1 as Rank 4 globally (mean |SHAP|=0.00439, direction −) confirms that burden escalation exhibits strong autoregressive structure. One-time or single-year interventions are structurally insufficient to break the escalation cycle: a household receiving a single year of emergency rental assistance will, absent continued support, return to the same structural conditions -stagnant renter income, sustained rent levels, low vacancy -that drove its original escalation. The plateau pattern observed in the borough-level forecasts -in which three of four boroughs stabilize after the first forecast step -confirms that the autoregressive lag structure converges rapidly under fixed macroeconomic assumptions, reinforcing the case for multi-year assistance commitments of minimum 2–3 years as a necessary design feature of effective burden-reduction programs.

5) Manhattan: Sub-Borough NTA-Level Targeting:

Manhattan's Rank 1 SHAP magnitude of 0.06891 (direction −) -the second lowest of any borough after the Bronx (0.05180) -reflects extreme intra-borough income heterogeneity rather than a fundamentally weaker affordability problem. Manhattan's near-zero mean residual (−0.000080) confirms the model is well-calibrated at the borough level, but borough-level policy instruments are poorly suited to Manhattan: programs calibrated to the borough mean will systematically under-serve high-burden NTAs in Washington Heights, East Harlem, Inwood, and Hamilton Heights while over-allocating resources to low-burden submarkets in Midtown and Lower Manhattan. Residential development linkage fees on luxury construction in low-burden Manhattan NTAs, structured to fund affordable unit production in high-burden NTAs, would create an intra-borough cross-subsidy mechanism that the current project-level inclusionary zoning framework does not provide.

6) From Predictive Model to Policy Administration:

The LightGBM model achieves sufficient predictive precision (Test R^2=0.9258, RMSE=0.03477) to function as an operational policy-targeting instrument. The critical administrative advantage over ACS-based monitoring is temporal: ACS 5-year estimates are released with an 18-to-24-month lag, meaning that by the time official burden statistics confirm neighborhood-level deterioration, one to two years of intervention opportunity have already elapsed. A model-based early-warning system, updated annually using the prior year's ACS release, would generate NTA-level burden forecasts 18 to 24 months ahead of official confirmation, allowing HPD, NYCHA, and the Mayor's Office to Prevent Homelessness to pre-position assistance resources in the specific NTAs the model identifies as crossing intervention thresholds at the ±0.03477 RMSE precision level confirmed on the held-out test set.

TABLE X: POLICY RECOMMENDATION MATRIX

Derived from SHAP Attribution, Borough-Level Forecasts, and Model Precision Findings (LightGBM, Test R^2=0.9258)

#	Empirical Finding	SHAP / Forecast Evidence	Borough	Policy Instrument	Agency	Urgency
1	rent_burden_30plus_pct is Rank 1 predictor in all four boroughs	Mean |SHAP| range: 0.05180 (Bronx) – 0.08632 (Brooklyn)	**All boroughs**	Expand NYC Emergency Rental Assistance Program (ERAP) eligibility to households at the 30% burden threshold, not restricted to acute crisis cases	HRA, HPD	**Critical**
2	rent_burden_30plus_pct is Rank 1 predictor in all four boroughs	Mean |SHAP| range: 0.05180 – 0.08632	**All boroughs**	Redesign CityFHEPS and Section 8 Housing Choice Voucher targeting to prioritize households in the 30–49% burden range before eviction filing occurs	HPD, HCR	**Critical**

#	Empirical Finding	SHAP / Forecast Evidence	Borough	Policy Instrument	Agency	Urgency
3	rent_burden_30plus_pct Rank 1 all boroughs; model precision sufficient for operational use	Test R^2=0.9258, RMSE=0.03477; ACS release lag 18–24 months	**All boroughs**	Deploy NTA-level LightGBM predictions as annual early-warning system, flagging NTAs crossing 30% burden share of 0.45 for proactive outreach ahead of ACS publication	HPD, DCP, Mayor's Office	**High**
4	Queens exhibits highest forecast burden growth and strongest temporal lag effect	Forecast +6.9% (2022–2025); Queens |SHAP| #1=0.07084 (−)	**Queens**	Targeted upzoning of transit corridors -Jackson Heights, Woodside, Astoria, Flushing - with mandatory inclusionary zoning set-asides in all new residential construction	DCP	**Critical**
5	Brooklyn exhibits second-highest forecast burden growth driven by sustained gentrification pressure	Forecast +4.3% (2022–2025); |SHAP| #1=0.08632 (+)	**Brooklyn**	Anti-displacement zoning and affordable unit set-asides in East New York, Crown Heights, Flatbush; operationalized through 'City of Yes for Housing Opportunity' rezonings	DCP, HPD	**Critical**
6	rent_to_income_ratio direction positive in Bronx only -no high-income absorptive capacity	SHAP direction: Bronx (−), Brooklyn (+), Manhattan (−), Queens (−)	**Bronx**	State-level EITC expansion targeting low-wage renter households, directly increasing the income denominator of the rent-to-income ratio	NYS Legislature, HRA	**High**
7	rent_to_income_ratio positive in Bronx only; elevated ratio translates directly to severe burden	Bronx forecast reaching 0.3146 by 2023, plateauing through 2025 (+4.9%)	**Bronx**	Index borough-level minimum wage adjustments to Bronx-specific rent growth rather than citywide averages, closing the lag of Bronx wage growth behind local rent appreciation	NYS DOL	**High**
8	Supply-side interventions alone insufficient in Bronx given income insufficiency structure	Bronx absolute burden highest (0.2998 in 2022); projected 0.3146 by 2025	**Bronx**	NYCHA public housing stock preservation and capital reinvestment in Bronx developments; replacement cost of equivalent affordable units through new construction is prohibitive	NYCHA, HUD, HPD	**Critical**
9	rent_burden_30plus_pct_lag1 is Rank 4 globally confirming autoregressive burden escalation	ean |SHAP|=0.00439 (−); formally: burden_30plus(i, t−1)	**All boroughs**	Mandate minimum 2–3 year rental assistance commitments calibrated to the one-year lag horizon; single-year emergency grants are structurally insufficient to break the escalation cycle	HRA, HUD	**High**
10	Temporal path dependence confirms households return to escalation conditions absent sustained support	rent_burden_30plus_pct_lag1 Rank 4 globally; mean |SHAP|=0.00439	**All boroughs**	Expand right-to-counsel (Local Law 136, 2019) to households at the 30% burden threshold facing lease non-renewal, not restricted to active eviction proceedings	NYS OCA, Legal Aid Society	**High**
11	Lease non-renewal and informal displacement precede eviction filing and are not captured in eviction records	Eviction rate ranked #9 globally in SHAP; lag structure implies pre-eviction pathway	**All boroughs**	Structured lease stabilization agreements with private landlords in high-lag NTAs via HPD property tax incentive programs, interrupting year-over-year rent escalation	HPD	**High**
12	Manhattan Rank 1 SHAP magnitude second lowest reflecting extreme intra-borough income heterogeneity	|SHAP| #1: Manhattan=0.06891 vs. Brooklyn=0.08632; rent_to_income_ratio direction (−)	**Manhattan**	Residential development linkage fees on luxury construction in low-burden Manhattan NTAs funding affordable units in Washington Heights, East Harlem, Inwood	DCP, HPD	**Moderate**
13	Manhattan borough-level programs systematically under-serve high-burden	Manhattan mean residual=−0.000080; near-zero confirms borough-level	**Manhattan**	NTA-level HPD proactive inspection prioritization and anti-harassment enforcement	HPD	**Moderate**

#	Empirical Finding	SHAP / Forecast Evidence	Borough	Policy Instrument	Agency	Urgency
	NTAs masked by low-burden averages	calibration masks NTA heterogeneity		in Washington Heights, East Harlem, Inwood, Hamilton Heights where model identifies rising moderate burden		
14	All boroughs show worsening forecast trajectory under baseline no-intervention scenario	Bronx +4.9%; Brooklyn +4.3%; Manhattan +2.8%; Queens +6.9% through 2025	**All boroughs**	Annual refresh of NTA-level LightGBM model using prior-year ACS 5-year estimates and updated ZORI values, producing updated NTA burden forecasts for NYC housing capital budget planning	HPD, DCP, OMB	**High**

Urgency: Critical = intervention required within 12 months given absolute burden level or forecast trajectory; High = intervention required within 1–3 years; Moderate = medium-term structural reform (3–5 years).
Risk Level classification: Based on absolute severe burden level in 2025 projection. Bronx classified Critical (0.3146, highest absolute burden; nearly 1 in 3 renter households severely cost-burdened). Queens and Brooklyn classified High (fastest relative growth rates of +6.9% and +4.3% respectively). Manhattan classified Moderate (lowest absolute burden 0.2148 and slowest growth +2.8%).
Agencies: HPD = NYC Dept. of Housing Preservation and Development; HRA = NYC Human Resources Administration; HCR = NYS Homes and Community Renewal; DCP = NYC Dept. of City Planning; NYCHA = NYC Housing Authority; HUD = U.S. Dept. of Housing and Urban Development; NYS OCA = New York State Office of Court Administration; OMB = NYC Office of Management and Budget; NYS DOL = New York State Dept. of Labor; Legal Aid Society = NYC nonprofit legal services provider.
Disclaimer: All policy recommendations are derived from empirical findings in this paper and represent the authors' analytical interpretations only. They do not constitute official policy positions of any cited agency. Causal inference cannot be drawn from the predictive model; recommendations reflect associations identified through SHAP attribution and forecast trajectories, not experimentally identified causal effects.

XII. Limitations and Future Work

1. ACS Sampling Uncertainty: ACS 5-year estimates carry margins of error (MOE) that can be substantial for small NTAs with fewer than 2,000 renter households. This analysis treats all point estimates as exact, without propagating MOE into prediction uncertainty intervals. Future work should incorporate ACS replicate weights or Monte Carlo resampling to produce uncertainty-aware burden forecasts.

2. Non-Causal Predictive Framework: The ensemble models are purely predictive and do not support causal inference. High SHAP importance does not imply that an intervention on a feature will produce a commensurate change in rent burden, due to confounding and reverse causality. Quasi-experimental designs -difference-in-differences around the 2019 Housing Stability and Tenant Protection Act (HSTPA), regression discontinuity at rezoning boundaries -are required for causal identification.

3. Borough-Level Spatial Analysis: Moran's I operates on only four borough-level units, severely limiting statistical power. Future work should apply NTA-level spatial weights using k-nearest-neighbor matrices or inverse-distance weights across all 239 NTAs, potentially revealing spatially clustered residuals not detectable at the borough level.

4. Static Forecast Assumptions: Borough-level forecasts rely on simplified constant annual growth rates under a single baseline scenario. Scenario analysis -optimistic: rent growth = income growth; adverse: rent growth = 5%/yr, income flat - would provide more policy-relevant uncertainty bounds for municipal planning purposes.

5. Missing Building-Level Features: The feature set excludes building-level variables that directly affect NTA-level affordability: rent stabilization unit counts, NYCHA public housing inventory, HPD housing quality complaint rates, and Certificate of Occupancy data. Incorporating NYC Department of Housing Preservation and Development registration data and the NYC Rent Guidelines Board's rent-stabilized apartment census represents a critical avenue for model improvement.

6. Data Quality and Outliers: The dataset contains extreme outlier values in vacancy_rate (max=1.000) and unemployment_rate (max=1.000) that likely reflect data quality artifacts in sparsely populated NTAs. Future work should apply winsorization or robust preprocessing to eliminate these outliers before modeling.

XIII. Conclusion

This paper presents the first NTA-level machine learning analysis of severe housing cost burden in New York City, employing a 2,512-observation panel dataset constructed from three complementary sources spanning 2012–2022. Under a rigorous temporal holdout evaluation protocol and 5-fold TimeSeriesSplit cross-validation, LightGBM achieves state-of-the-art predictive performance: Test R^2=0.9258, RMSE=0.03477, MAE=0.01919, Mean CV R^2=0.865 ± 0.045, with the smallest train-test generalization gap of any benchmarked model (ΔR^2=0.017).

SHAP analysis establishes that rental market dynamics -the 30%-threshold burden rate (mean |SHAP|=0.06806), vacancy rate (mean |SHAP|=0.01474), and rent-to-income ratio (mean |SHAP|=0.00804) -dominate predictive attribution in every borough, with the rental market feature group accounting for ΔR^2=0.0668 (7.22%) in the ablation study -the largest contribution of any group by a margin of ΔR^2=0.0220 over the second-ranked spatial identifiers group. The borough-level SHAP decomposition reveals that severe rent burden in NYC is governed by a structurally uniform escalation mechanism: the same three features rank as the top three predictors in every borough without exception, with Brooklyn as the consistent directional outlier across all three top features -recording positive directions on rent_burden_30plus_pct and rent_to_income_ratio where all other boroughs record negative directions -reflecting the distinct gentrification dynamics driving simultaneous escalation across both burden thresholds in that borough.

The Moran's I spatial diagnostic (I=−0.0801, p=0.0829 n.s., permutation test n_perm=9,999) provides reassurance that LightGBM's economic features capture borough-level spatial structure adequately, with three of four boroughs showing slight over-prediction (Bronx −0.009840, Manhattan −0.000080, Queens −0.004880) and only Brooklyn showing marginal under-prediction (+0.001212). Baseline forecasts project continued deterioration across all four boroughs through 2025: Queens (+6.9%, from 0.2256 to 0.2411) faces the highest relative burden growth, while the Bronx remains the most severely burdened in absolute terms (+4.9%, reaching 0.3146 by 2023 and plateauing through 2025 -meaning nearly one in three renter households severely cost-burdened). These projections underscore the urgency of scaling the NYC Emergency Rental Assistance Program, accelerating affordable unit production through 'City of Yes' zoning reforms, and strengthening income-side interventions in the Bronx where supply-side responses alone are structurally insufficient.

Three priorities for future research emerge: (1) NTA-level spatial modeling with k-nearest-neighbor weights across all 239 NTAs to resolve the limited statistical power of the n=4 borough-level Moran's I analysis; (2) integration of rent stabilization status, NYCHA inventory, and HPD complaint data as additional features to address the building-level omitted variable gap identified in Section XII; and (3) causal difference-in-differences designs around the 2019 Housing Stability and Tenant Protection Act to move beyond predictive attribution toward experimentally identified policy effect estimates. The 14-instrument policy recommendation matrix anchored to actual SHAP attribution values, borough-level residual patterns, and forecast trajectories constitutes the primary translational contribution of this analysis, providing a direct bridge between NTA-level machine learning findings and operationally specific housing policy administration.

Acknowledgments

The author gratefully acknowledges the U.S. Census Bureau for providing open access to ACS 5-year microdata through the Census Data API, and the NYC Department of City Planning for publishing the official census tract-to-NTA geographic crosswalk. Eviction records are made available by the NYC Office of Court Administration through NYC Open Data. Zillow Observed Rent Index data is provided by Zillow Research. This research was conducted independently and received no external funding. The author declares no conflicts of interest.

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